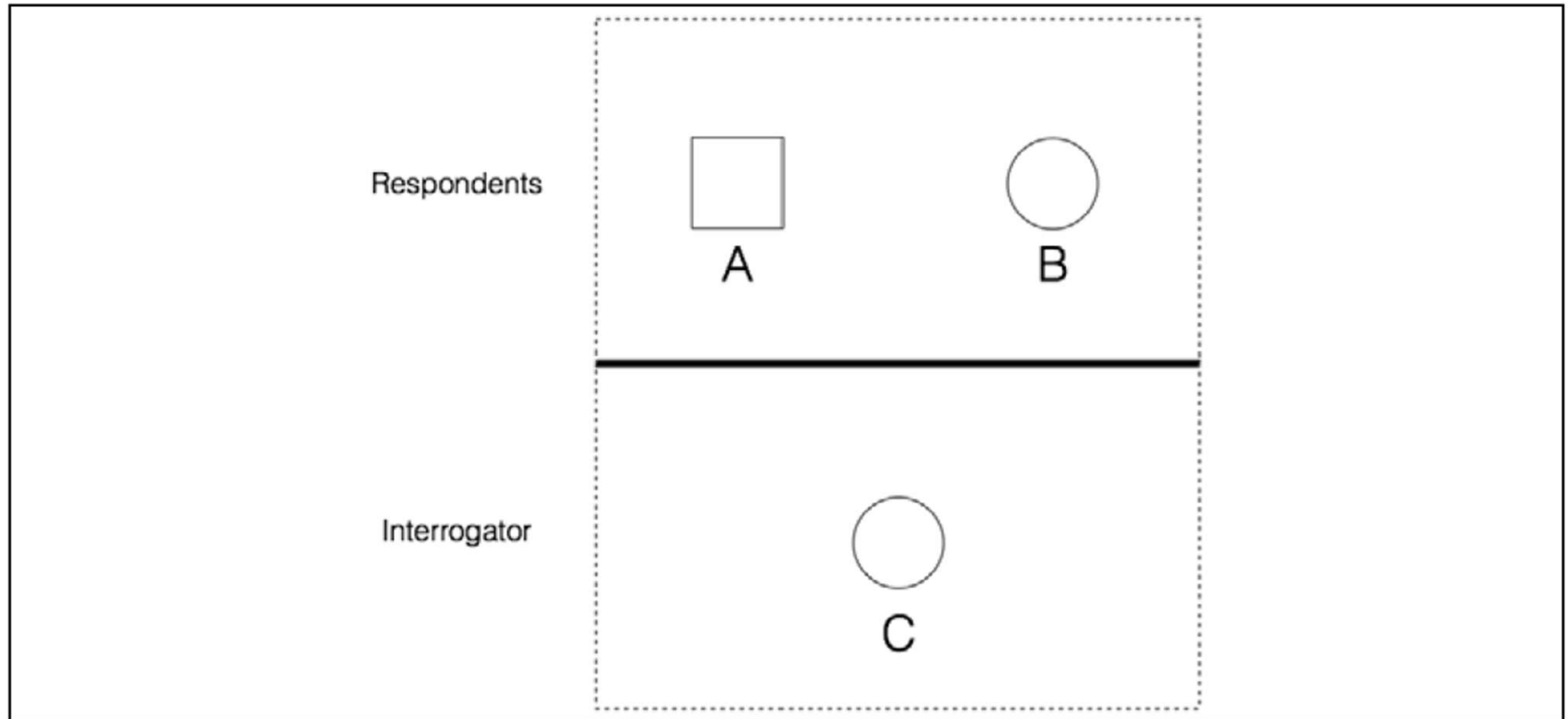


Lecturer

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-

Artificial Intelligence

Turing Test



- **Natural language processing:** The machine needs this to communicate with the interrogator. The machine needs to parse the sentence, extract the context, and give an appropriate answer.

❖ **Knowledge representation:**

- The machine needs to store the information provided before the interrogation.
 - It also needs to keep track of the information being provided during the conversation so that it can respond appropriately if it comes up again.
-

- **Reasoning:** It's important for the machine to understand how to interpret the information that gets stored.

❖ **Machine learning:** This is needed so that the machine can adapt to new conditions in real time. The machine needs to analyze and detect patterns so that it can draw inferences.

Artificial Intelligence

AI involves techniques that equip computers to emulate human behavior, enabling them to learn, make decisions, recognize patterns, and solve complex problems in a manner akin to human intelligence.

Machine Learning

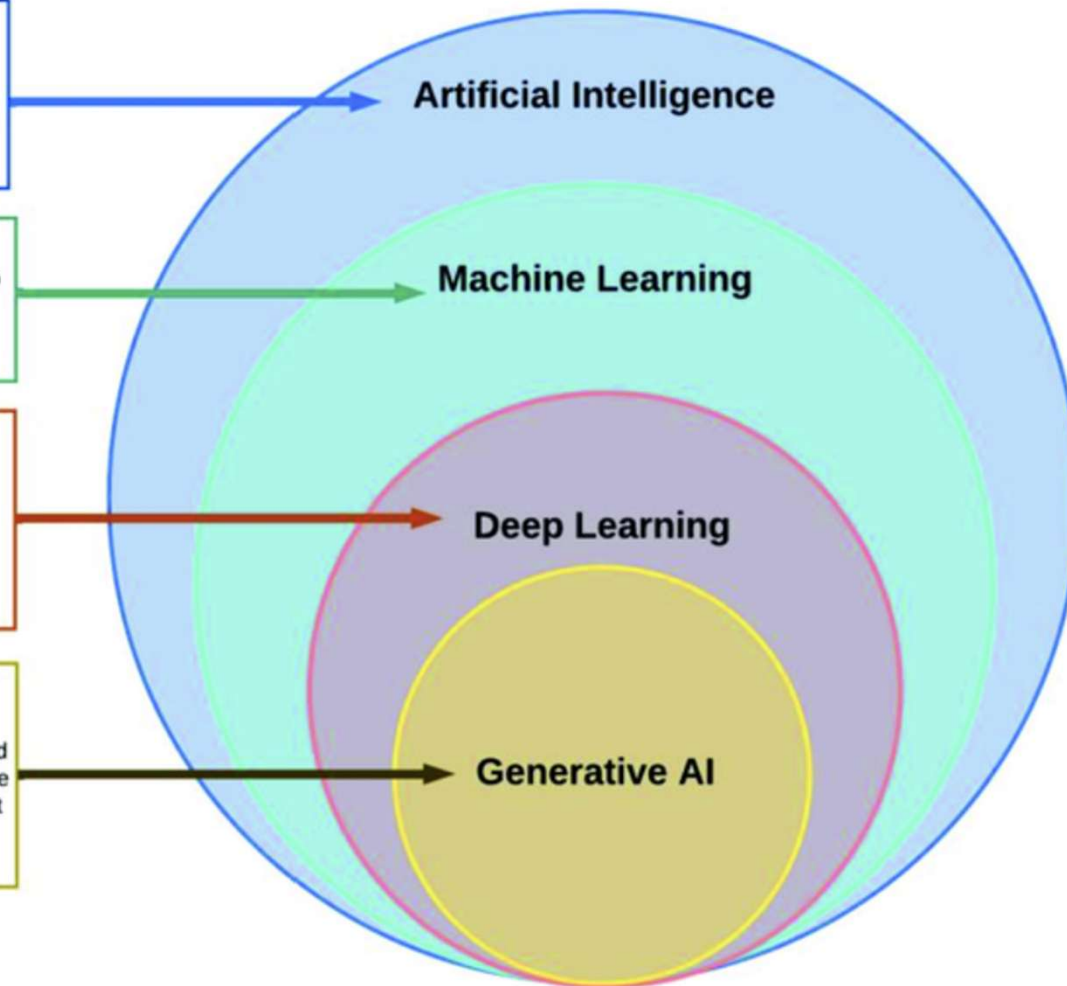
ML is a subset of AI, uses advanced algorithms to detect patterns in large data sets, allowing machines to learn and adapt. ML algorithms use supervised or unsupervised learning methods.

Deep Learning

DL is a subset of ML which uses neural networks for in-depth data processing and analytical tasks. DL leverages multiple layers of artificial neural networks to extract high-level features from raw input data, simulating the way human brains perceive and understand the world.

Generative AI

Generative AI is a subset of DL models that generates content like text, images, or code based on provided input. Trained on vast data sets, these models detect patterns and create outputs without explicit instruction, using a mix of supervised and unsupervised learning.

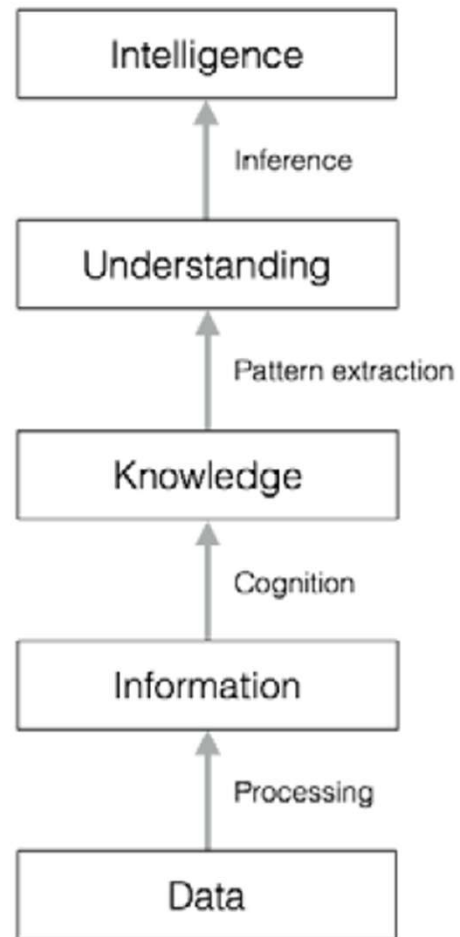


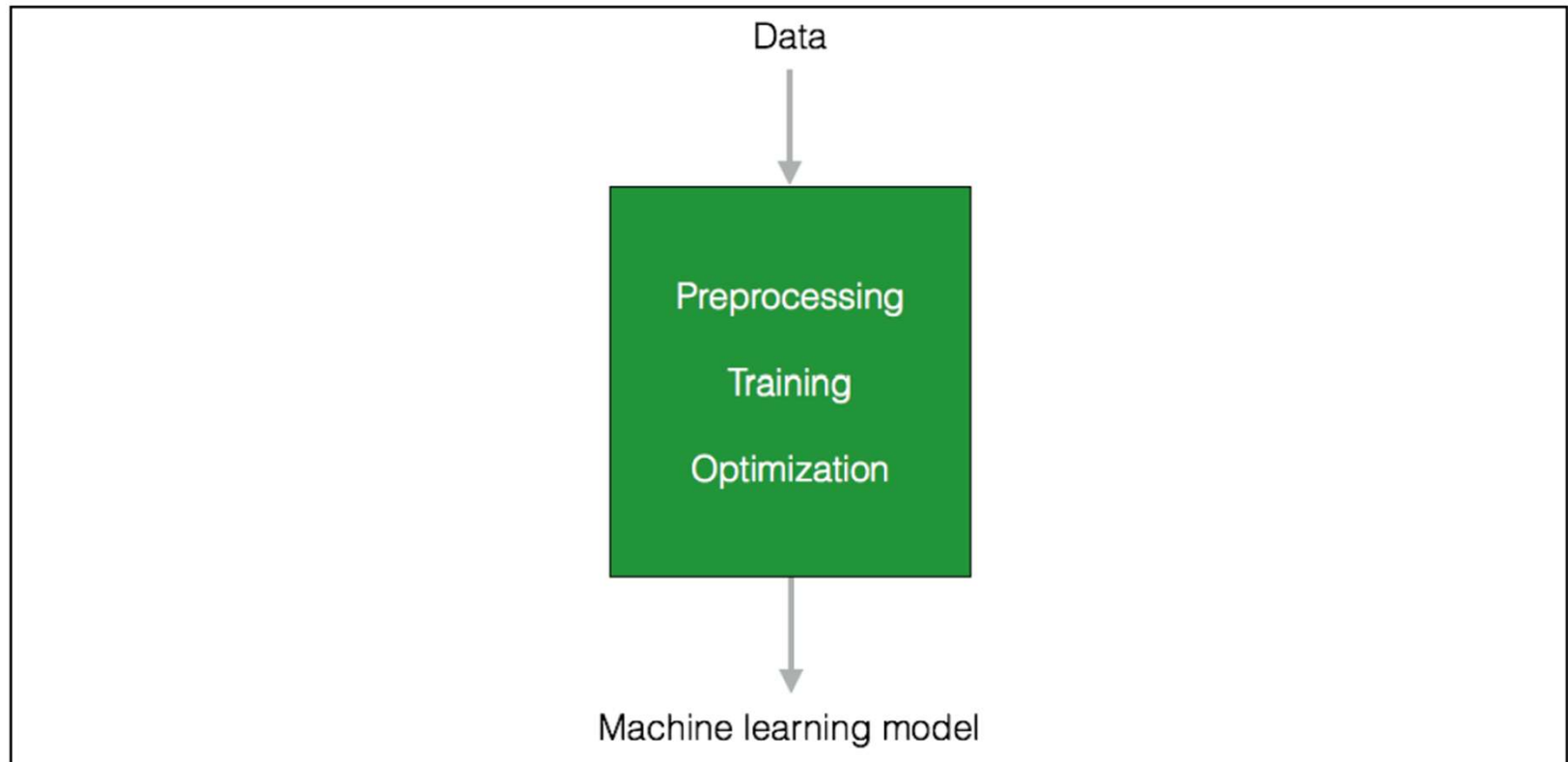
what is “Intelligence?”

❖ The area of computer science that studies how machines can perform tasks that would normally require a sentient agent

- ❖ the area of computer science that studies how machines can closely imitate human
- ❖ intelligence

- ❖ • What is AI and why do we need to study it?
 - ❖ • What are some applications of AI?
 - ❖ • A classification of AI branches
 - ❖ • The five tribes of machine learning
 - ❖ • What is the Turing test?
 - ❖ • What are rational agents?
 - ❖ • What are General Problem Solvers?
 - ❖ • How to build an intelligent agent
-

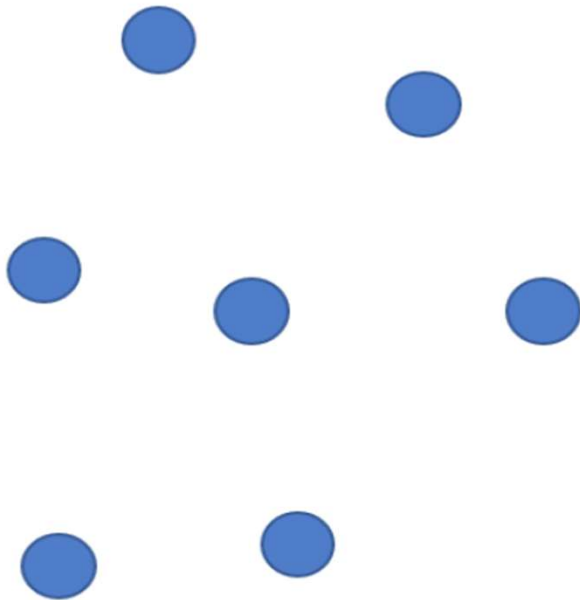




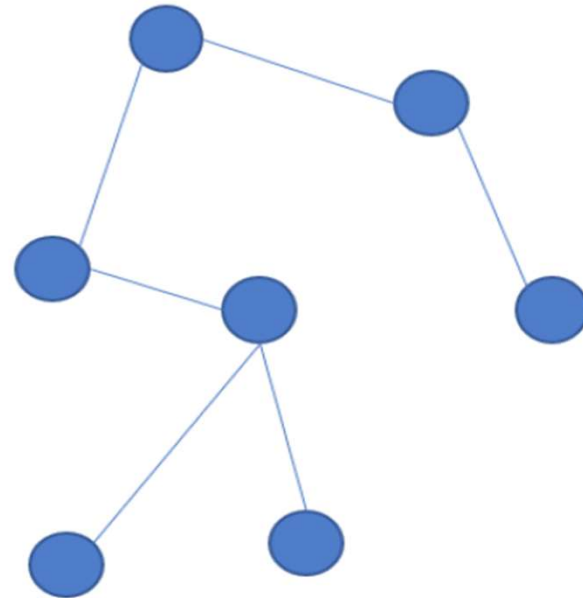
Branches of AI

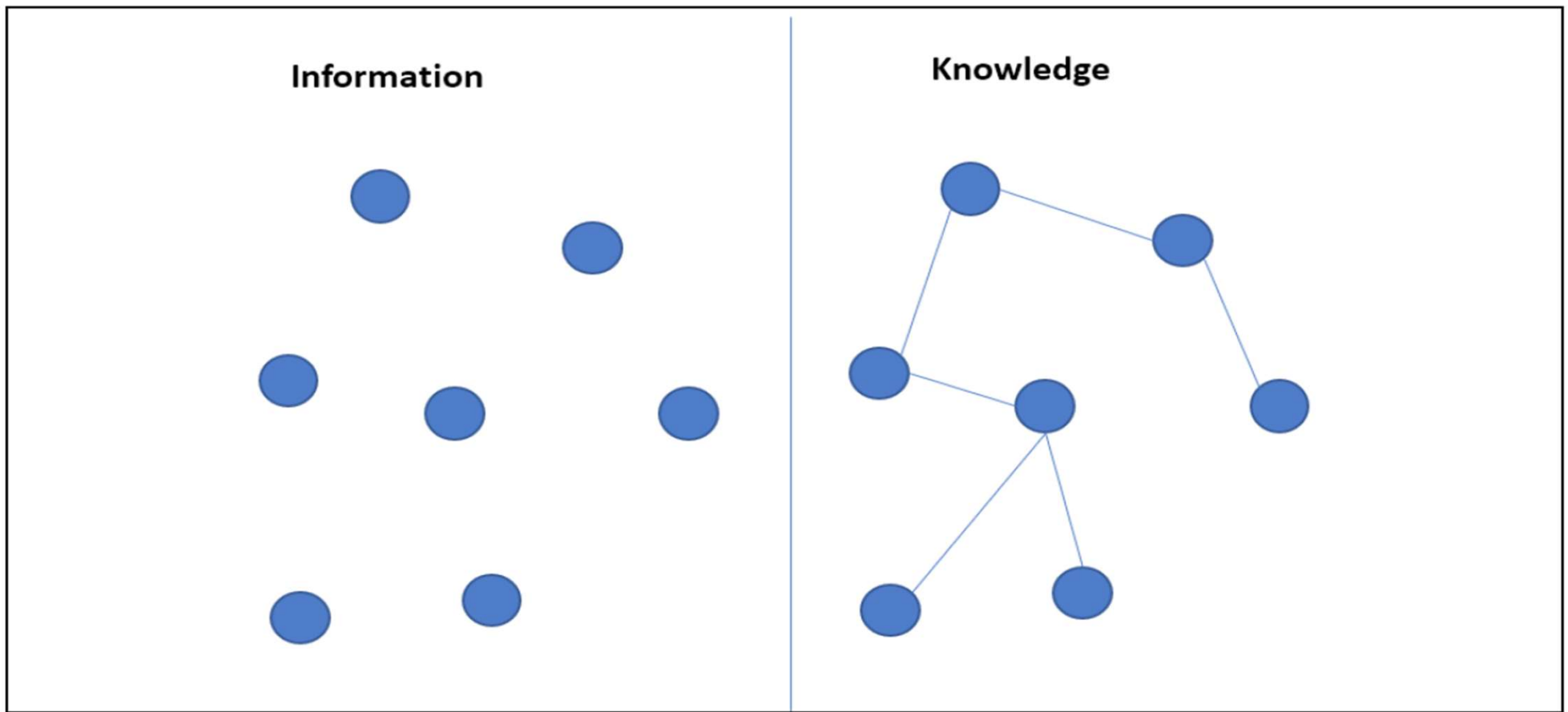
- ❖ Machine learning and pattern recognition
 - ❖ Logic-based AI
 - ❖ Search
 - ❖ Knowledge representation
 - ❖ Planning
 - ❖ Heuristics
 - ❖ Genetic programming
-

Information

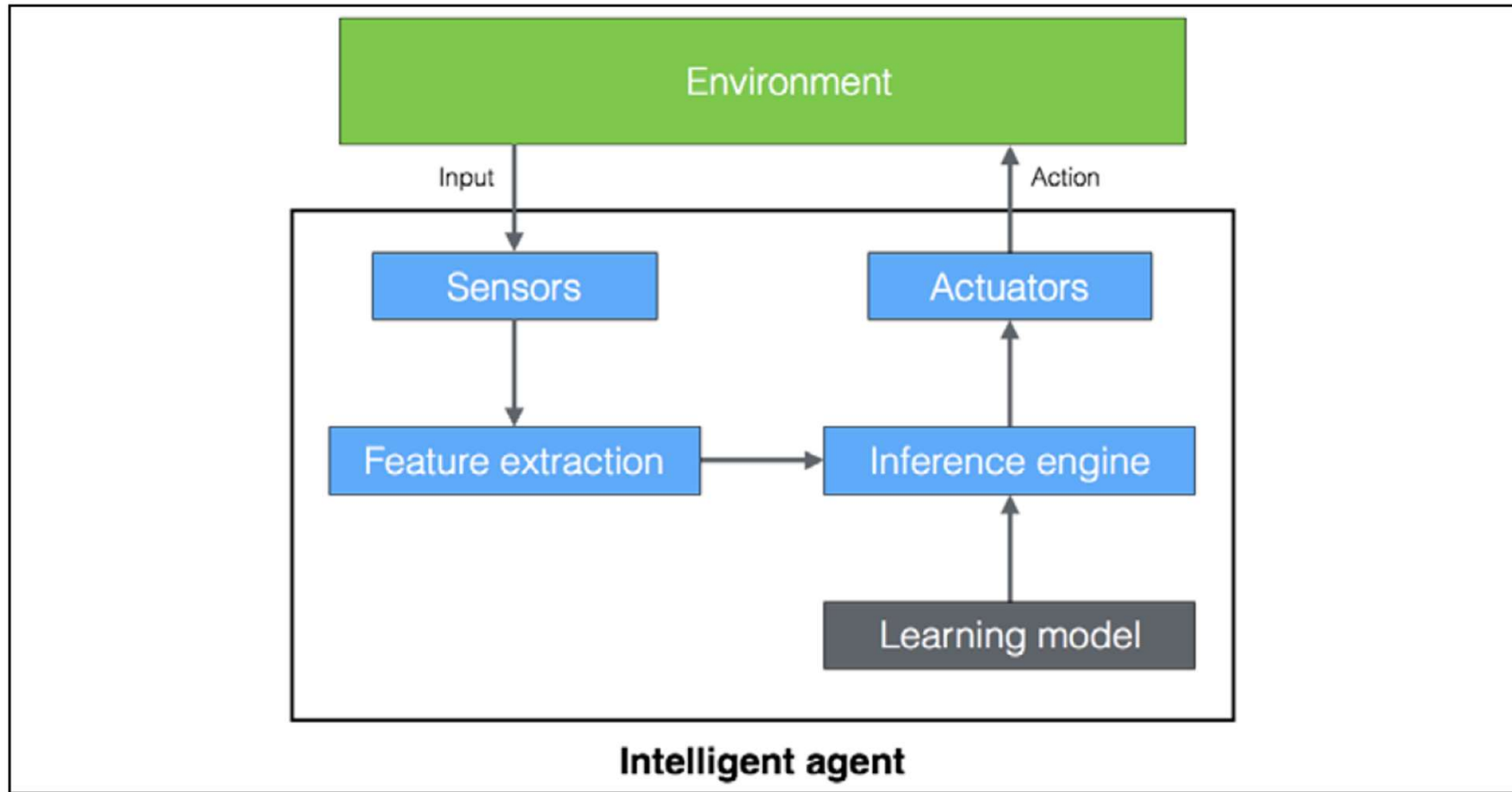


Knowledge





- ❖ The facts about the world around us need to be represented in some way for a system to make sense of them.
- ❖ If knowledge is represented efficiently, systems can be smarter and more intelligent



With machine learning, we want to program our machines to use labeled data to solve a given problem by learning how to **extract patterns** and relationships.

Five tribes of machine learning

Tribe	Origins	Dominant algorithm	Proponents
Symbolists	Logic and Philosophy	Inverse deduction	Tom Mitchell Steve Muggleton Ross Quinlan
Connectionists	Neuroscience	Backpropagation	Yan LeCun Geoffrey Hinton Yoshua Bengio
Evolutionaries	Biology	Genetic programming	John Koza John Holland Hod Lipson
Bayesians	Statistics	Probabilistic inference	David Heckerman Judea Pearl Michael Jordan
Analogizers	Psychology	Kernel machines	Peter Hart Vladimir Vapnik Douglas Hofstadter

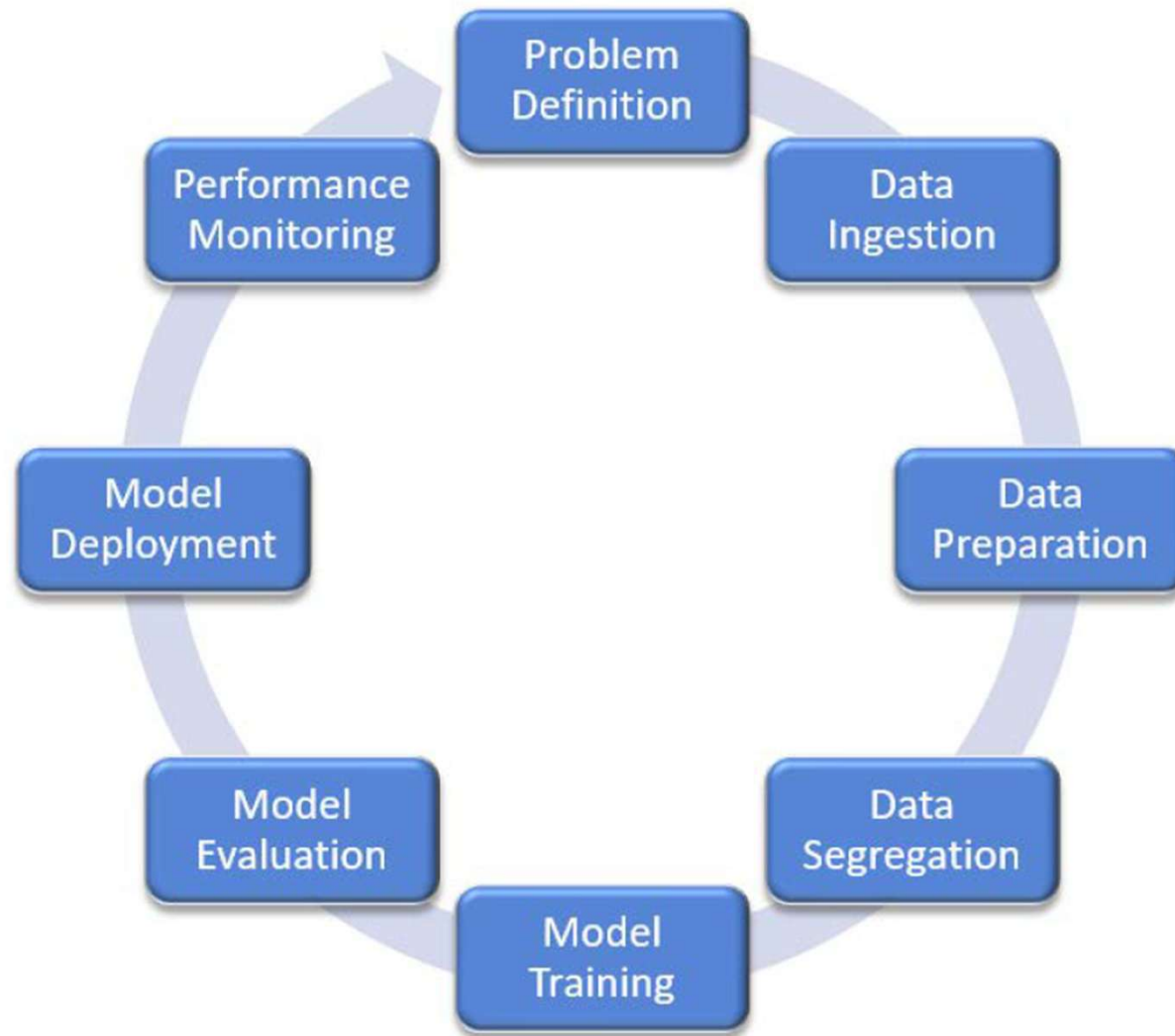
Applications

- ❖ Natural Language Processing (NLP):
Google Assistant, Amazon Alexa, Apple Siri
 - ❖ Generative AI: chatGPT
 - ❖ Automation: Google's Waymo
 - ❖ Smart home
 - ❖ Games
 - ❖ Recommendation systems: E-commerce
 - ❖ Making movies, photos, slides, ...
-

All these assistants will allow you to perform all or at least most of these tasks:

- Control devices in your home
 - Play music and display videos on command
 - Set timers and reminders
 - Make appointments
 - Send text and email messages
 - Make phone calls
 - Open applications
 - Read notifications
 - Perform translations
 - Order from e-commerce sites
-

Machine Learning Pipelines



Machine Learning Pipelines

The main steps in a machine learning pipeline are:

- 1. Problem Definition:** Define the business problem

- 2. Data Ingestion:** Identify and collect the dataset.

3. Data Preparation: Process and prepare the data using techniques :

- Impute missing values
 - Remove duplicate records
 - Normalize values (change numeric values in a dataset to use a common scale)
 - Perform another type of cleanup or mappings
 - Complete feature extraction
 - Eliminate correlated features
 - Perform feature engineering
-

4. Data Segregation: Split the data into a training set, validation set, and testing set.

5. Model Training: Train the machine models against the training dataset. This is the core of data science.

6. Candidate Model Evaluation: Measure the performance of the models using test and validation subsets of data to determine model accuracy.

7. Model Deployment: Once a model is chosen, deploy it into production for inference.
